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OSI Model - 7 Layers



What is OSI Model?

The **OSI Model (Open Systems Interconnection)** is a **7-layer framework** that explains how data travels from one system to another.

Think of it as a **step-by-step journey of data**.



OSI Model - 7 Layers



7 Layers of OSI Model

Layer No.	Layer Name	Example
7	Application	HTTP, FTP
6	Presentation	Encryption
5	Session	Session management
4	Transport	TCP/UDP
3	Network	IP
2	Data Link	MAC
1	Physical	Cables



7. Application Layer

The application layer is the topmost layer of the OSI model and is closest to the end user. This layer provides the interface for users and software applications to interact with the network. It handles high-level functions like file transfers, email, and web browsing.

What it does:

User interacts here.

Examples:

- Web browsing
- Email
- File transfer

Protocols:

- HTTP / HTTPS
- FTP
- SMTP

SOC Perspective:

- Phishing attacks
- Malicious URLs
- Suspicious API calls



6. Presentation Layer

The presentation layer is responsible for the translation, encryption, and compression of data.

It ensures that data sent from the application layer is in a format that can be understood by the receiving system.

This layer often handles tasks like data formatting, character encoding (e.g., ASCII to Unicode), and data encryption.

What it does:

- Data formatting
- Encryption / Decryption

Examples:

- SSL/TLS
- Encoding

SOC Perspective:

- Encryption analysis
- SSL stripping (MITM)



5. Session Layer

The session layer establishes, manages, and terminates sessions between two communicating devices.

It ensures that data exchange is synchronized and allows applications to communicate effectively without interference.

What it does:

- Maintains session between systems

Example:

- Login session
- Session cookies

SOC Perspective:

- Session hijacking
- Token reuse



4. Transport Layer

The transport layer ensures the reliable delivery of data between two devices across a network.

It segments the data received from the upper layers and provides error detection and correction.

Additionally, it controls flow to prevent congestion and guarantees that data is delivered in the correct order.

What it does:

- End-to-end communication
- Reliability

Protocols:

- TCP (reliable)
- UDP (fast, unreliable)

SOC Perspective:

- Port scanning
- SYN flood attacks
- Suspicious ports



3. Network Layer

The network layer is responsible for determining the best path to send data across multiple networks.

This layer handles logical addressing and routing, enabling data to travel from one network to another, regardless of the physical infrastructure.

Routers operate at this layer, forwarding data between different networks based on IP addresses.

What it does:

- Routing
- IP addressing

Protocol:

- IP (IPv4 / IPv6)

SOC Perspective:

- IP tracking
- Geolocation
- Suspicious external connections



2. Data Link Layer

The data link layer sits above the physical layer and is responsible for the reliable transfer of data across a physical network link.

This layer ensures that data is error-free and properly framed for transmission.

It also manages how devices on the same local network communicate with each other.

What it does:

- MAC address communication
- Local network delivery

SOC Perspective:

- ARP spoofing (MITM)
- MAC anomalies



1. Physical Layer

The physical layer is the lowest layer of the OSI model, and it deals with the actual transmission of raw binary data (1s and 0s) over a physical medium, such as cables, fiber optics, or wireless signals.

It defines the hardware elements involved in data transfer, such as network interfaces, cables, switches, and the electrical signals used to carry the data.

What it does:

- Hardware transmission
- Electrical signals

SOC Perspective:

- Rarely used directly
- But important in hardware attacks



How does communication happen in the OSI Model

Communication in the OSI Model follows a systematic process called **Encapsulation and Decapsulation**:

1. **Sender's Side:**

- Data is generated at the application layer and moves downward through the layers.
- Each layer adds its own header, and sometimes a footer, encapsulating the data with layer-specific information (e.g., addressing, error checks).

2. **Transmission:**

- The encapsulated data (now called a packet) is transmitted over the physical medium to the receiving device.

3. **Receiver's Side:**

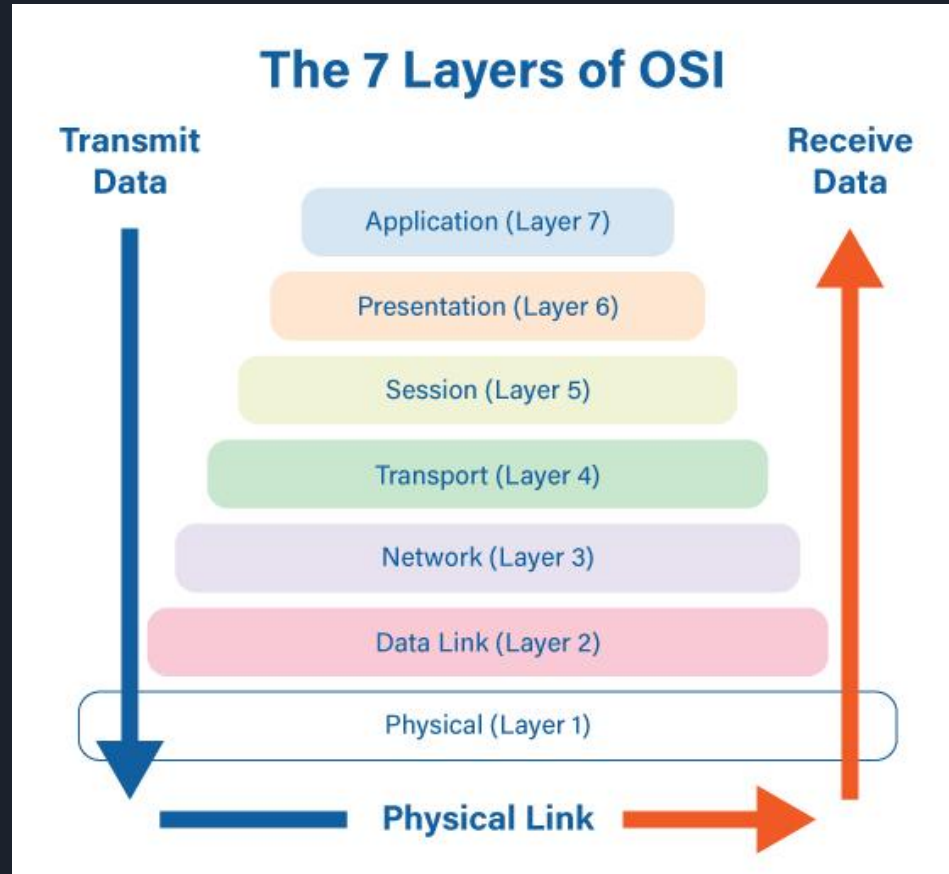
- The process is reversed as data moves upward through the OSI layers.
- Each layer removes its corresponding header/footer (decapsulation) until the original data is presented to the receiving application.

Note : This structured approach ensures data integrity and reliability, even over complex networks.

OSI Model - 7 Layers



Encapsulation & Decapsulation

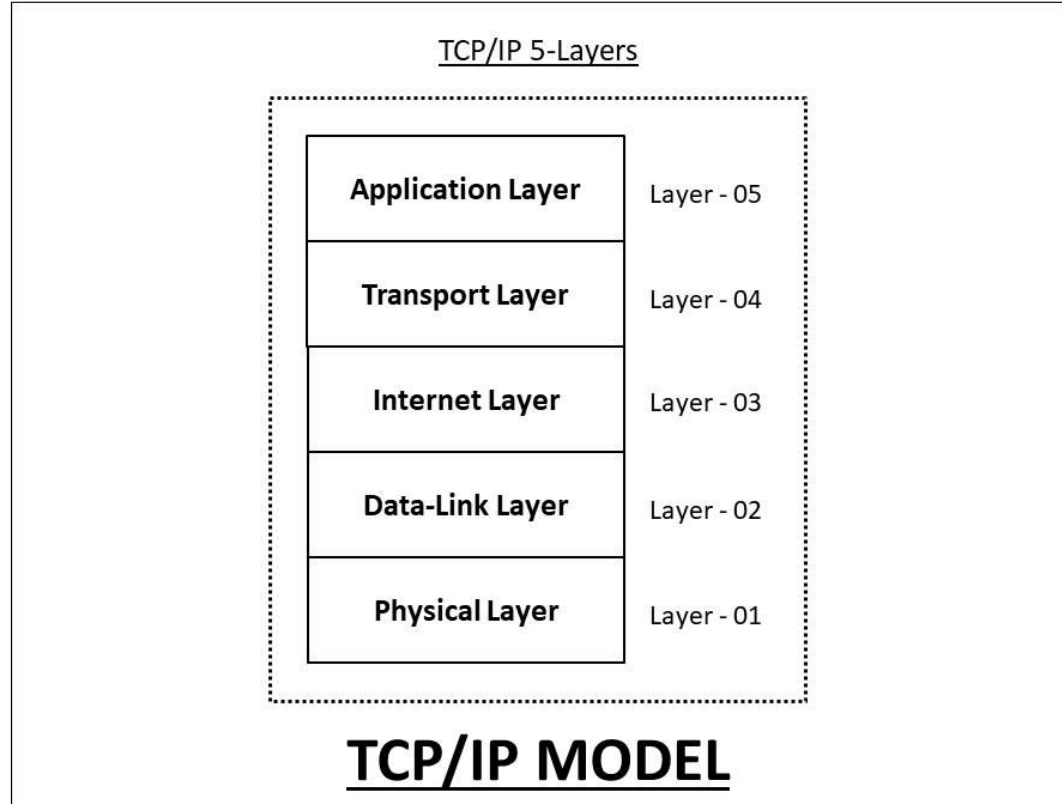




What is TCP/IP Model?

Used in real networks (Internet).

Simplified version of OSI.





TCP/IP Layers

TCP/IP Layer	Maps to OSI
Application	OSI 7,6,5
Transport	OSI 4
Internet	OSI 3
Network Access	OSI 2,1



1. Application Layer

Includes:

- HTTP
- HTTPS
- DNS
- FTP

SOC View:

- Phishing
- DNS attacks
- API abuse



2. Transport Layer

Protocols:

- TCP
- UDP

SOC View:

- Port scanning
- SYN flood
- Traffic anomalies



3. Internet Layer

Protocol:

- IP

SOC View:

- Suspicious IP
- External communication
- Botnet traffic



4. Network Access Layer

Includes:

- MAC
- ARP
- Ethernet

SOC View:

- ARP spoofing
- Internal attacks



OSI vs TCP/IP (Important for Interviews)

Feature	OSI	TCP/IP
Layers	7	4
Use	Conceptual	Practical
Real usage	Less	High



SOC Example (Mapping)

Alert:

Suspicious login via HTTPS from unknown IP

Analysis:

- Application → HTTP login
- Transport → Port 443
- Network → External IP
- Data Link → Internal network



SOC Analyst Thinking

Whenever you see a log, break it like:

Which layer is involved?

Example:

- Port issue → Transport
- IP issue → Network
- DNS issue → Application
- ARP issue → Data Link

TCP/IP Model (Real-World Model)



Save for later

Created by:
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OSI Model vs TCP/IP Model

